Ari Khoudary

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Education

University of California, Irvine
Ph.D., Cognitive Sciences — Summer 2026 (anticipated)
M.S., Cognitive Neuroscience — Fall 2024
Affiliate, Center for Neurobiology of Learning and Memory (CNLM) — Fall 2021-present
Affiliate, Center for Theoretical Behavioral Sciences (CTBS) — Winter 2023-present
Advisors: Dr. Aaron Bornstein and Dr. Megan Peters

Boston College
B.S., Psychology (honors) and Philosophy, *magna cum laude* — May 2019
Scholar of the College Thesis: *Securing the Reliability of Episodic Memory*
Advisors: Dr. Maureen Ritchey (psychology) and Dr. Richard Atkins (philosophy)

Research Experience

Research Assistant & Lab Manager, De Brigard Lab, Duke University — Summer 2019-Summer 2021
Undergraduate Research Fellow, Ritchey Lab, Boston College — Fall 2017-Spring 2019
NSF-REU Fellow, Carrasco Lab, New York University — Summer 2018

Peer-reviewed Publications *(* denotes equal contribution)*

NB: I began publishing with my chosen name (Ari) in August 2022

Khoudary, A., Bornstein, A.M.*, Peters, M.A.K.* (2025) Investigating implicit and explicit expectations in perceptual decision making. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society* [pdf]

Khoudary, A., Peters, M.A.K.*, Bornstein, A.M.* (2025) Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model. *European Journal of Neuroscience.* https://onlinelibrary.wiley.com/doi/full/10.1111/ejn.70098 [pdf]

Miceli, K.*, Morales-Torres, R.*, **Khoudary, A.**, Faul, L., Parikh, N., & De Brigard, F. (2023). Perceived plausibility modulates hippocampal activity in episodic counterfactual thinking. *Hippocampus*. https://doi.org/10.1002/hipo.23583 [pdf] [code]

Khoudary, A., Peters, M.A.K.*, Bornstein, A.M.* (2022) Precision-weighted evidence integration predicts time-varying influence of memory on perceptual decisions. *Proceedings of the 2022 Conference on Cognitive Computational Neuroscience*. [pdf] [code]

Khoudary, A., Hanna, E., O'Neill, K.G., Iyengar, V., Clifford, S., De Brigard, F.*, Cabeza, R.*, Sinnott-Armstrong, W.* (2022). A Functional Neuroimaging Investigation of Moral Foundations Theory. *Social Neuroscience*, 1-17. https://doi.org/10.1080/17470919.2022.2148737 [pdf] [code]

Khoudary, A., O'Neill, K., Faul, L., Murray, S., Smallman, R., De Brigard, F. (2022). Neural differences between internal and external episodic counterfactual thoughts. *Philosophical Transactions of the Royal Society B*. https://doi.org/10.1098/rstb.2021.0337 [pdf] [code]

De Brigard, F., **Khoudary, M**., & Murray, S. (2022). Times Imagined and Remembered. In Hoerl, C., McCormack, T., & Fernandes, A. (Eds.). *Temporal Asymmetries in Philosophy and Psychology.* Oxford University Press. [pdf] [code]

Mele, A., Nadelhoffer, T., **Khoudary, M.** (2021). Folk psychology and proximal intentions, *Philosophical Psychology,* 34:6, 761-783, DOI: 10.1080/09515089.2021.1915471 [pdf] [code]

Manuscripts in Preparation

Khoudary, A, Peters, M. A. K., Bornstein, A.M. (in prep) Precision-weighted integration explains dynamic effects of memory on perceptual decisions.

Khoudary, A., Bornstein A.M.*, Peters, M. A. K.* (in prep) Saving the baby from the bathwater: key considerations for principled use of large language models in cognitive neuroscience.

Khoudary, A., Leonard, B.T., Yoo, J., Bornstein, A.M. (in prep) The indispensability of normative models for understanding individual differences in psychopathology.

Invited Talks

Philosophical tools for evaluating computational models of cognition — October 2025
Poldrack Lab at Stanford University; invited by Dr. Russell Poldrack

Uncertainty resolution explains dynamic effects of memory on perceptual decisions — October 2025
Aly Lab at The University of California, Berkeley; invited by Dr. Mariam Aly

Uncertainty resolution explains dynamic effects of memory on perceptual decisions — May 2025
MesoLab at Indiana University Bloomington; invited by Dr. Rob Goldstone

Reasoning goals & representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model — October 2024
Cognitive Sciences Colloquium at UC Irvine; invited by Dr. Aaron Bornstein

Balancing principles & practice in computational cognitive neuroscience: a case study from bounded evidence accumulation — January 2024
Cognition and Individual Differences Lab at UC Irvine; invited by Dr. Joachim Vandekerckhove

Conference Presentations *(* denotes equal contribution; ^ indicates author mentored by me)*

Khoudary, A, Bornstein, A.M.*, Peters, M. A. K.* (2025) Philosophical tools for evaluating computational models of cognition. *Poster at the Society for Neuroscience (Theme K)*

Khoudary, A., Peters, M. A. K.*, Bornstein, A.M.* (2025) Precision-weighted integration explains dynamic effects of expectations on perceptual decisions. *Poster at the Society for Neuroscience.*

Chandler, L.^, **Khoudary, A.**, Peters, M. A. K.*, Bornstein, A.M.* (2025) Experience to Expectation: How

Learning Dynamics Shape Memory-Guided Perceptual Decision-Making. *Poster at the 2025 Summer Institute in Neuroscience Symposium.*

Khoudary, A., Chandler, L.^, Peters, M. A. K.*, Bornstein, A.M.* (2025) Investigating interactions between learned and reported expectations in perceptual decision making. *Poster at the 47th Annual Meeting of the Cognitive Science Society.*

Khoudary, A., Peters, M. A. K.*, Bornstein, A.M.* (2025) A principled model of uncertainty resolution explains dynamic effects of expectations on perceptual decisions. *Talk at the 2025 meeting of The Society for Philosophy and Neuroscience; Talk at the 2025 Bay Area Memory Meeting.*

Khoudary, A. (2024) Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from bounded evidence accumulation. *Talk at the 7th annual meeting of the Deep South Philosophy & Neuroscience workgroup; Poster at the 2024 meeting of the Philosophy of Science Association.*

Jackson, N.^, **Khoudary, A.**, Peters, M. A. K.*, Bornstein, A.M.* (2024) Testing a new theory of perceptual decision-making. *Poster at the 2024 Summer Institute in Neuroscience Symposium*.

Khoudary, A., Bornstein, A.M.*, Peters, M. A. K.* (2024) Explaining dynamic effects of memory on perceptual decisions. *Data Blitz talk at the 2024 CNLM Spring Meeting.*

Khoudary, A., Bornstein, A.M.*, Peters, M. A. K.* (2023) Characterizing dynamic effects of memory on perceptual evidence accumulation. *Poster at the RIKEN CBS Summer Program Poster Session.*

Khoudary, A., Bornstein, A.M.*, Peters, M. A. K.* (2023) Perceptual decisions result from dynamic precision-weighted integration of memory and visual information. *Talk at the 26th Annual Meeting of the Association for the Scientific Study of Consciousness (ASSC)*.

Khoudary, A., Peters, M. A. K.*, Bornstein, A.M* (2023) Characterizing time-varying effects of memory on perceptual decisions. *Poster at the 2023 International Conference on Learning and Memory.*

Miceli, K.^, Morales-Torres, R, **Khoudary, A.**, Faul, L., Parikh, N., De Brigard, F. (2023) Perceived plausibility modulates hippocampal activity in episodic counterfactual thinking. *Poster at the 2023 Cognitive Neuroscience Society meeting.*

Khoudary, A., Peters, M. A. K.*, Bornstein, A.M.* (2022). Precision-weighted evidence integration predicts time-varying influence of memory on perceptual decisions. *Poster at the 2022 Cognitive Computational Neuroscience (CCN) meeting.*

Khoudary, M., O'Neill, K.G., Faul, L., Murray, S., Smallman, R., De Brigard, F. (2021; 2022). Neural differences between internal and external episodic counterfactual thoughts. *Poster presented virtually at the 2021 Neuromatch Conference (NMC). Poster at the 2022 Cognitive Neuroscience Society (CNS) Annual Meeting.*

Chen, J.*, Elowsky, C.*, **Khoudary, M**.*, Revsine, C.* (2020) Predicting fMRI Responses: a Machine Learning Approach. *Neuromatch Academy project presented in fulfillment of the Interactive Track.*

Khoudary, M., Hanna, E., O'Neill, K.G., Iyengar, V., Clifford, S., De Brigard, F., Cabeza, R., Sinnott-Armstrong, W. (2020; 2021). A Functional Neuroimaging Investigation of Moral Foundations

Theory. *Poster at the virtual Cognitive Neuroscience Society (CNS) Annual Meeting. Poster at the virtual 2021 meeting of the Society for Philosophy and Psychology (SPP).*

Khoudary, M., Cooper, R., Ritchey, M. (2019). Effects of divided attention on item and source memory. *Poster at the Boston College Psychology Undergraduate Research Conference (PURC).*

Khoudary, M. (2019). How attention alters perception. *Talk at Nu Rho Psi outreach event.*

Khoudary, M., Jigo, M., Carrasco, M. (2018). Characterizing the effects of covert exogenous attention on contrast sensitivity across the visual field. *Poster at the NYU Summer Student Research Conference. Talk at the Center for Neural Science Summer Undergraduate Research Program Symposium.*

Awards

Travel Award, UCI Center for Theoretical Behavioral Sciences	Fall 2025
Travel Award, UCI Association of Graduate Students	Spring 2025
Schneiderman-NIMH T32, UCI CNLM	Spring 2023-Spring 2025
Jared M. Roberts Memorial Award, UCI CNLM	Spring 2023
DECADE Outstanding Student Representative Award, University of California, Irvine	Spring 2023
Graduate Research Fellowship Honorable Mention, National Science Foundation	Spring 2023
DECADE Professional Development Award, University of California, Irvine	Winter 2022
Diversity Recruitment Fellowship, University of California, Irvine	Spring 2021
Research Experience for Undergraduates Fellow, National Science Foundation	Summer 2018

Academic Honors

Scholar of the College, Boston College	Spring 2019
Phi Beta Kappa, Boston College	Spring 2019
Order of the Cross and Crown, Boston College	Spring 2019
Dean's List, First Honors	Spring 2016, Spring 2017, Fall 2017, Spring 2018, Fall 2018, Spring 2019
Nu Rho Psi, National Honor Society in Neuroscience	Fall 2018
Psi Chi, National Honor Society in Psychology	Fall 2017
Honors Program, Boston College Psychology Department	Fall 2017
Honors Program, Boston College Morrissey College of Arts and Sciences	Spring 2015

Selected Leadership & Outreach

Colloquium Committee, UC Irvine Cognitive Sciences Department	Spring 2024-present
Mentor, Summer Institute in Neuroscience, UCI CNLM	Summers 2024 & 2025
Certificate in Mentorship Excellence, UCI Graduate Division	Fall 2023
Representative, Diverse Educational Community & Doctoral Experience, UC Irvine	Fall 2022-Fall 2025
Founder, Philosophy and Neuroscience Journal Club, Duke University	Spring 2020-Spring 2021
Mentor, Psychology Department Mentorship Program, Boston College	Fall 2018-Spring 2019
President and Co-Founder, Nu Rho Psi, Boston College	Spring 2018-Spring 2019
President, Students for Education Reform, Boston College	Fall 2016-Spring 2018

Mentorship Experience

Lauryn Chandler, Howard University undergraduate Summer 2025

- Summer Institute in Neuroscience project: individual differences in learning & decision-making

Aidan Goeschel, UC Irvine undergraduate Spring 2025-present

- Ongoing project: evaluating and refining LLM-based metrics of narrative flow

Nurazmera Mossa, UC Irvine undergraduate Fall 2024-present

- Ongoing project: behavioral modeling of confidence in memory-guided perceptual decision-making

Nadeja Jackson, Howard University undergraduate Summer 2024

- Summer Institute in Neuroscience project: piloting a novel model-based perceptual decision task

Professional Experience

Teaching Assistant, Memory, Prof. Aaron Bornstein Spring 2023
Teaching Assistant, Intro to Human Memory, Prof. Christine Lofgren Winter 2023
Teaching Assistant, Neurobiology of Cognition, Prof. Alyssa Brewer Fall 2022
Teaching Assistant, Honors Research Methods, Profs. Aaron Bornstein & Nadia Chernyak Spring 2022
Teaching Assistant, Intro to Human Memory, Prof. Christine Lofgren Winter 2022
Teaching Assistant, Sensation and Perception, Prof. Virginia Richards Fall 2021
Program Coordinator, Summer Seminars in Neuroscience and Philosophy Summer 2019-Summer 2021

Reviewing

Cerebral Cortex, *Memory and Cognition*, *Journal of Experimental Psychology: General, Open Mind, Philosophical Psychology*

Skills

Programming and Software
Advanced proficiency: R; MATLAB
Intermediate proficiency: Python; Git; JAGS; PLS and PsychToolbox for MATLAB; PsychoPy
Novice proficiency: JASP; TensorFlow; EEGLAB toolbox for MATLAB
Data Collection (human subjects)

fMRI (8 coil 3T GE MR750)	EEG (BioSemi 64 channel gel)
Behavioral (Qualtrics, Amazon Mechanical Turk, Prolific)	Eye Tracking (EyeLink 1000)

References

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